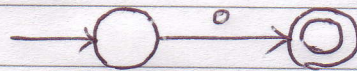


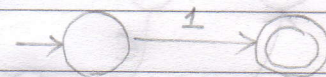
FOUNDATION OF COMPUTING THEORY - HOMEWORK 04

Q.1. $R = (01)^* \cup (00 \cup 1)^*$

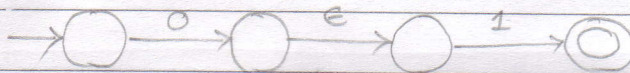
$R_1 = 0$



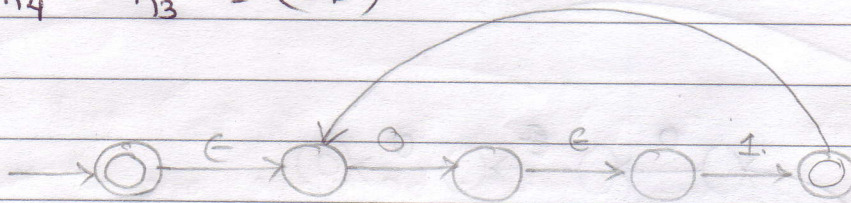
$R_2 = 1$



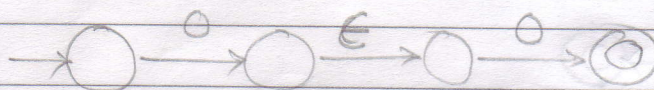
$R_3 = R_1 \circ R_2 = 01$



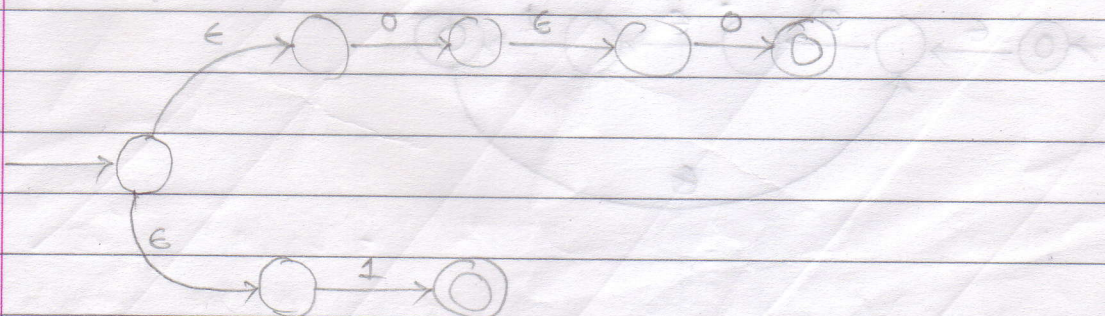
$R_4 = R_3^* = (01)^*$



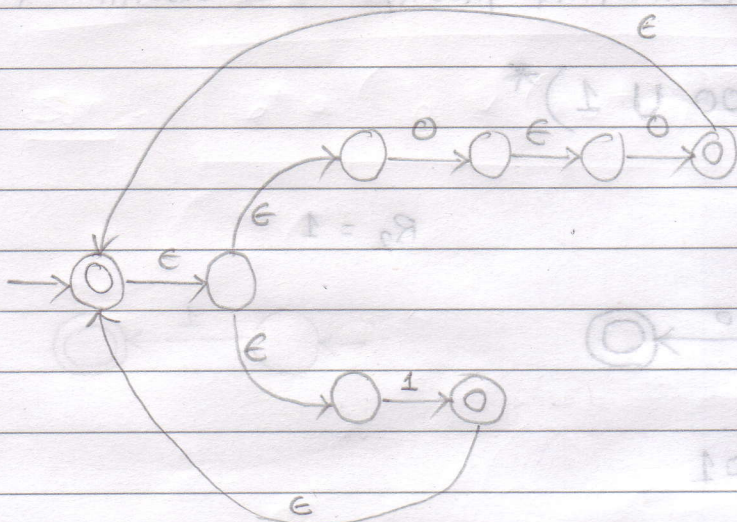
$R_5 = R_1 \circ R_1 = 00$



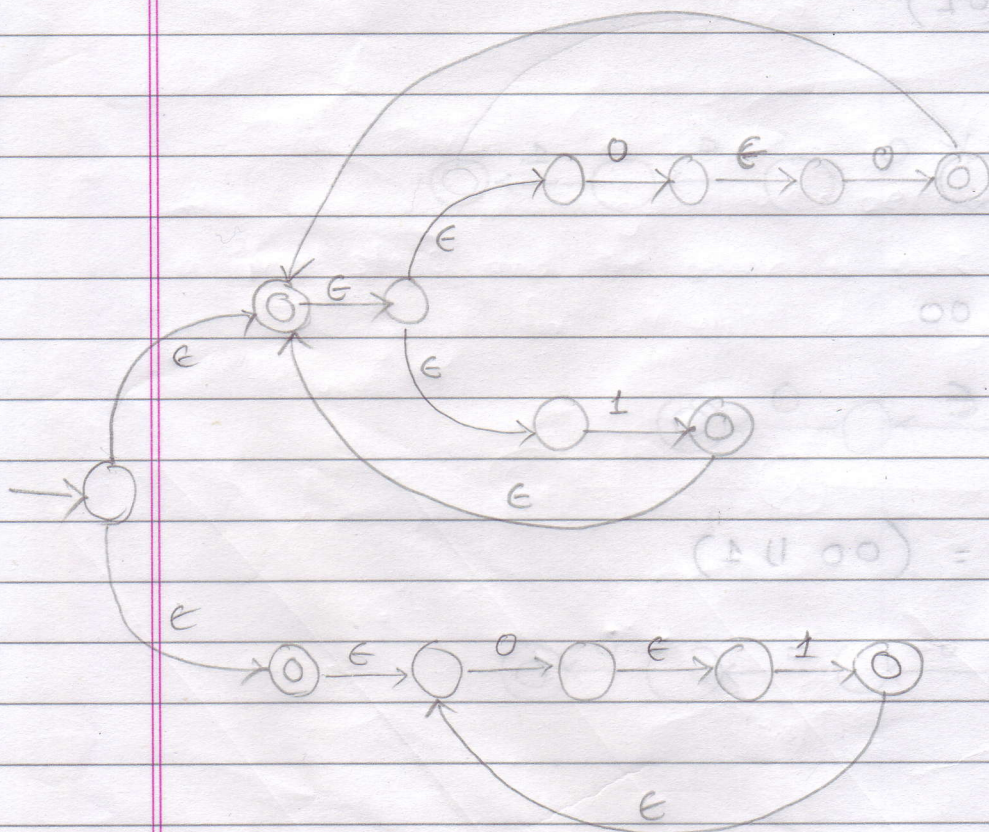
$R_6 = R_5 \cup R_2 = (00 \cup 1)$



$$R_7 = R_6^* = (00 \cup 1)^*$$



$$R_8 = R_4 \cup R_7 = (01)^* \cup (00 \cup 1)^* = R$$

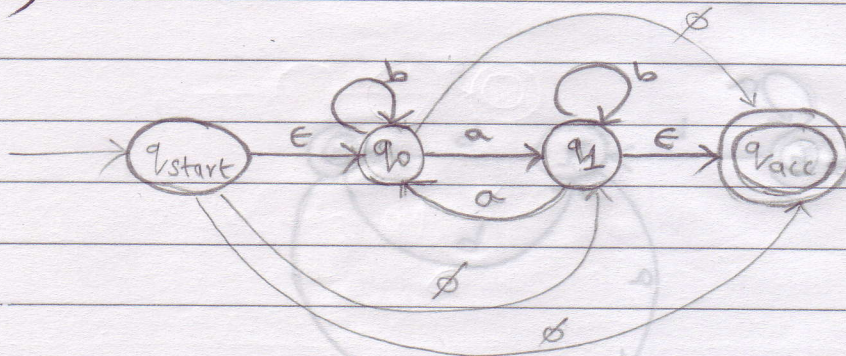


Q.2. $M = (\{q_0, q_1\}, \{a, b\}, \delta, q_0, \{q_1\})$

$$\delta(q_i, a) = q_{1-i} \quad i \in \{0, 1\}$$

$$\delta(q_i, b) = q_i \quad i \in \{0, 1\}$$

a)



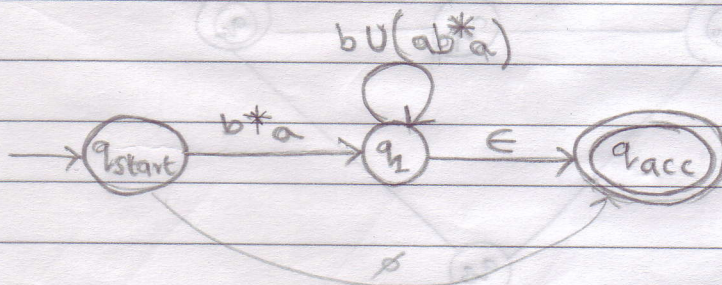
b) $q_{rip} = q_0$

$$\delta(q_1, q_1) = b \cup (ab^*a) = b \cup (ab^*a)$$

$$\delta(q_1, q_{acc}) = \epsilon \cup (ab^*\emptyset) = \epsilon$$

$$\delta(q_{start}, q_1) = \emptyset \cup (\epsilon b^*a) = b^*a$$

$$\delta(q_{start}, q_{acc}) = \emptyset \cup (\epsilon b^*\emptyset) = \emptyset$$



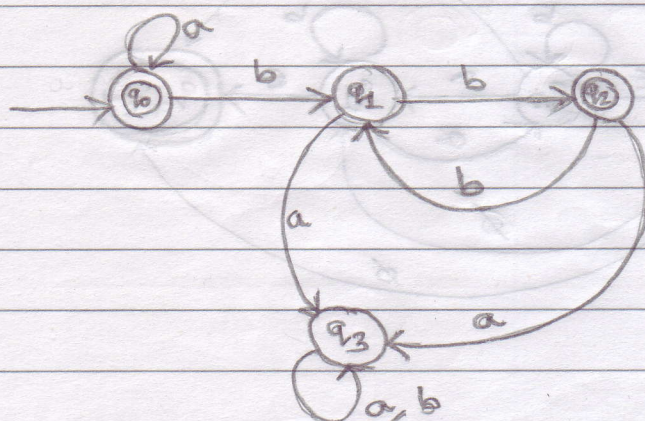
c) $q_{rip} = q_1$

$$\delta(q_{start}, q_{acc}) = \emptyset \cup (b^*a(b \cup (ab^*a))^*\epsilon)$$

Soln:- $b^* a (b^*)$

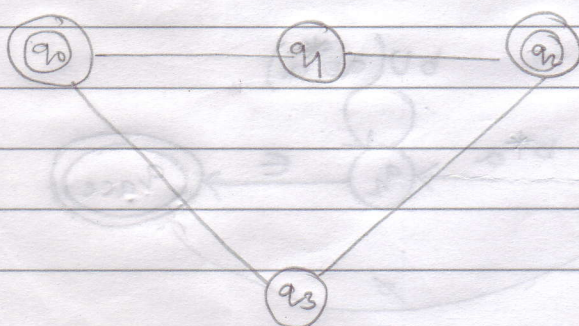
Soln:- $b^* a (b \cup (ab^* a))^*$

Q. 3

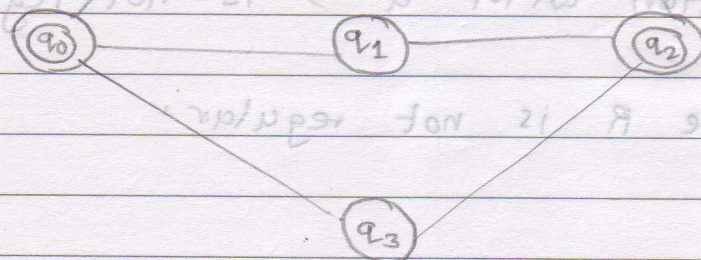


The pairs (q_0, q_1) , (q_0, q_3) , (q_1, q_2) and (q_3, q_2) are pairwise distinguishable because one of them is a final state and the other is not.

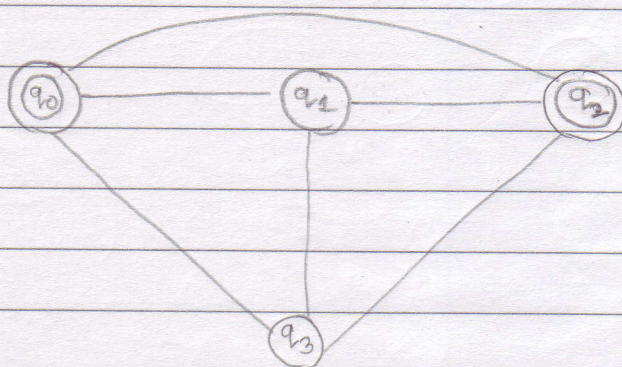
The graph for pair-wise distinguishable states is as follows:



Now, $\delta(q_0, a) = q_0$ & $\delta(q_2, a) = q_3$
And (q_0, q_3) is already an edge in the graph. So add (q_0, q_2) to the graph.
The new graph looks as follows



Now, $\delta(q_1, b) = q_2$ and $\delta(q_3, b) = q_2$
 But And (q_1, q_3) is already in the graph. So add (q_1, q_3) to the graph.
 The new graph looks as follows:-



Since all the states are pair-wise distinguishable, the DFA is a minimum DFA for the language $a^*(bb)^*$

Q.4. $R = \{a^i b^k \mid i \neq k\}$

Since $i \neq k$, we have 2 cases

i) $k > i$

then we can write $a^i b^k$ as $(a^i b^i) \cdot b^{(k-i)}$

Now $a^i b^i$ is not regular and hence its concatenation with $b^{(k-i)}$ is also not regular.

ii) $k < i$

then we can write $a^i b^k$ as $a^{(k-i)} \cdot (a^k b^k)$

Q.4. $R = \{a^i b^k \mid i \neq k\}$

- Let us assume that R is not regular.
- Since we have $i \neq k$, we have 2 cases.

i) $k > i$

- if $k > i$, we can write $a^i b^k$ as $(a^i b^i) \circ b^{(k-i)}$
 - ~~Now concatenation is closed under~~
 - Now regular languages are closed under concatenation.
- So for R ~~to~~ to be regular, both $a^i b^i$ and $b^{(k-i)}$ should be regular. But we know that $a^i b^i$ is not regular.
- Hence our assumption is wrong.
 - Hence R is not regular for $k > i$.

ii) $k < i$

- if $k < i$, we can write $a^i b^k$ as ~~$a^k b^k$~~
 $a^{(i-k)} \circ (a^k b^k)$.
 - Now regular languages are closed under concatenation.
- So for ~~if~~ R to be regular, both $a^{(i-k)}$ and $a^k b^k$ should be regular. But we know that $a^k b^k$ is not regular.
- Hence our assumption is wrong.
 - Hence R is not regular for $k < i$.

- Therefore, R is not regular.